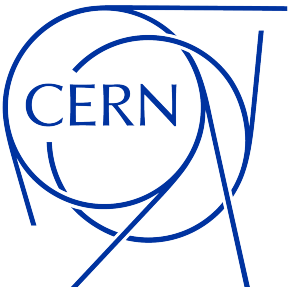


First Steps in Geant4: Examples and Documentation

Geant4-11.1 reference - Based on previous Geant4 courses

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Covered topics

This lesson is about taking the first steps in Geant4, it shows how to sniff around *examples and documentation*

- ◆ **Geant4 installation**

- ◆ **Examples**

- ✿ Build and examine `exampleB1`
- ✿ Basic examples
- ✿ Extended examples
- ✿ Advanced examples

- ◆ **Documentation**

- ✿ Installation Guide
- ✿ Documentation for Application/Toolkit Developers

- ◆ **User support**

Step #1: Geant4 website

- ◆ Where to find our documentation, download the code, access the user forum, inspect future events, ...

The screenshot shows the Geant4 website homepage. At the top is a navigation bar with links: About, Download, Documentation, User Forum, Bug Reports, Events, and Contact Us. The main header features the Geant4 logo and a large banner with the text: "Geant4 Toolkit for the simulation of the passage of particles through matter. Its areas of application include high energy, nuclear and accelerator physics, as well as studies in medical and space science." A blue button labeled "Getting started" is positioned on the right side of the banner. Below the banner, there are four main sections: "Get started" with a power icon and text "Everything you need to get started with Geant4. I'm ready to start!"; "Download" with a download icon and text "Geant4 source code and installers are available for download, with source code under an open source license. Latest: 11.1.2"; "Docs" with a document icon and text "Documentation for Geant4, along with tutorials and guides, are available online. Read documentation"; and "News" with a newspaper icon and text "30 Jun 2023 Release 11.2.beta, 19 Jun 2023 Release 11.1.2, 23 Mar 2023 2023 Planned Features, 03 Mar 2023 Release 11.0.4, 10 Feb 2023 Release 11.1.1". Below the "Get started" section is a grid of six small images showing various particle simulation results. Below the "Download" section is a group photo of the Geant4 collaboration team, with the word "Collaboration" written below it. Below the "Docs" section is a snippet of C++ code for a Geant4 macro.

Step #2: Install Geant4

The [installation guide](https://geant4-userdoc.web.cern.ch/UsersGuides/InstallationGuide/html/) is available at <https://geant4-userdoc.web.cern.ch/UsersGuides/InstallationGuide/html/>

- ◆ Refer to the guide for installation *dependencies*, *supported platforms* and *cmake options*

Example of Geant4 installation:

- ◆ Download source code from <https://geant4.web.cern.ch/support/download> or from [github](#), then

Bash - Example of Geant4 (latest patch) installation

```
$ unzip geant4-11.1.2.zip
$ mkdir geant4-11.1.2-build; cd geant4-11.1.2-build
$ cmake -DCMAKE_INSTALL_PREFIX=/path-to-install/geant4-11.1.2-install
-DGEANT4_INSTALL_DATA=ON -DGEANT4_USE_QT=ON -DGEANT4_BUILD_MULTITHREADED=ON
/path-to/geant4-11.1.2/
$ make -j6
$ make install
```

Selected *cmake* options:

- download and install data libraries (see lessons on physics)
- use qt for visualization
- build geant4 with multithreaded capability

Step #3: Build examples

(1/2)

Examples (geant4/examples/) are useful applications to learn Geant4 features (from basic to advanced)

- ◆ Building example B1, the first basic example

Bash - Building example B1

```
$ source /path-to/geant4-11.1.2-install/bin/geant4.sh

$ cmake -DGeant4_DIR=/path-to/geant4-11.1.2-install/lib/Geant4-11.1.2/
/path-to/geant4-11.1.2/examples/basic/B1/

$ make

$ ls

CMakeCache.txt    Makefile    exampleB1    exampleB1.out    run1.mac    vis.mac    CMakeFiles
cmake_install.cmake    exampleB1.in    init_vis.mac    run2.mac
```

\$./exampleB1

Several files are created when you compile an example.
Look for the executable file and the *.mac macro card setting some parameters.

Step #3: Build examples

(2/2)

Examples (geant4/examples/) are useful applications to learn Geant4 features (from basic to advanced)

◆ Building example B1, the first basic example

Bash - Building example B1

```
$ source /path-to/geant4-11.1.2-install/bin/
```

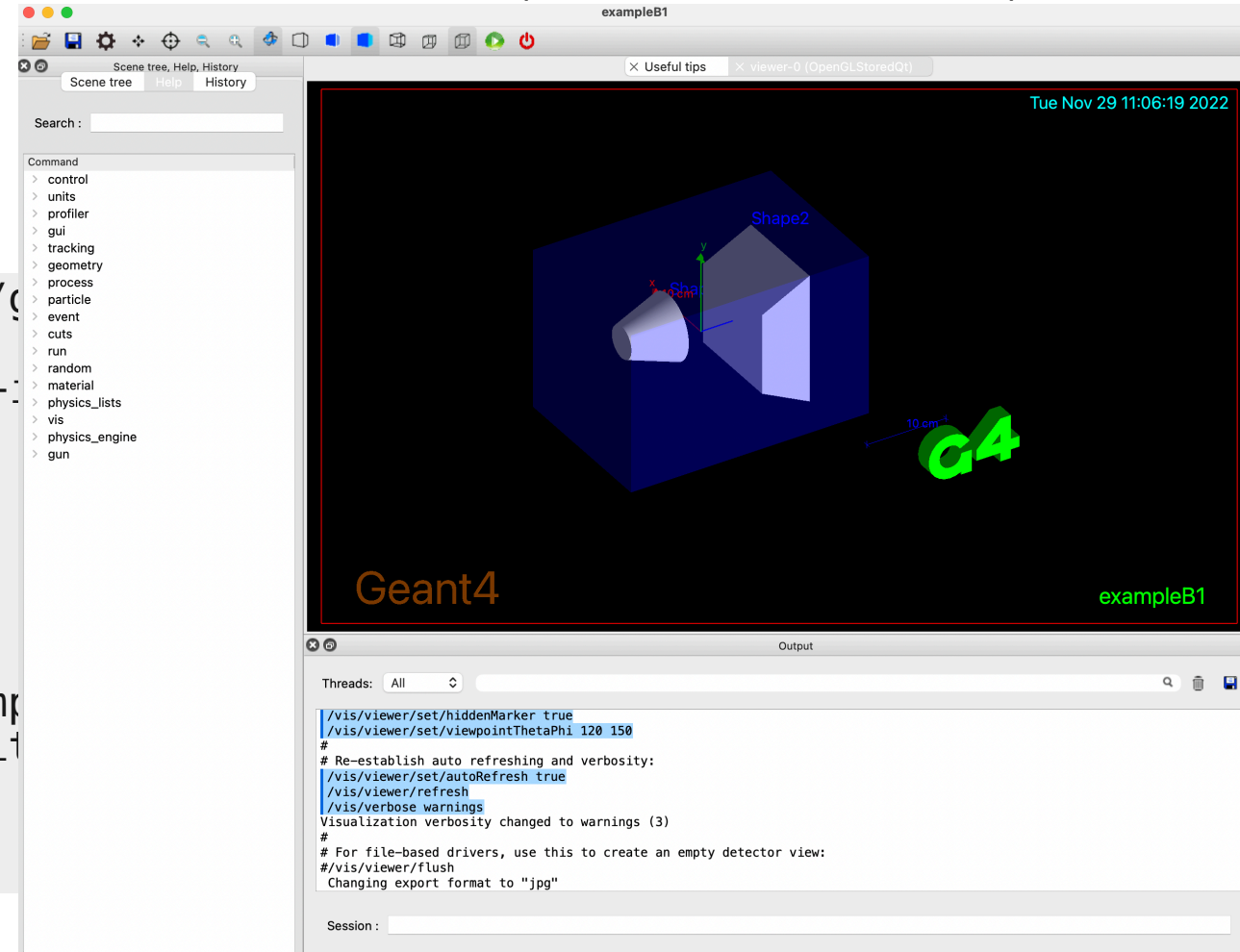
```
$ cmake -DGeant4_DIR=/path-to/geant4-11.1.2-  
/path-to/geant4-11.1.2/examples/basic/B1/
```

```
$ make
```

```
$ ls
```

```
CMakeCache.txt      Makefile      exampleB1      exam  
cmake_install.cmake      exampleB1.in      init
```

```
$ ./exampleB1
```

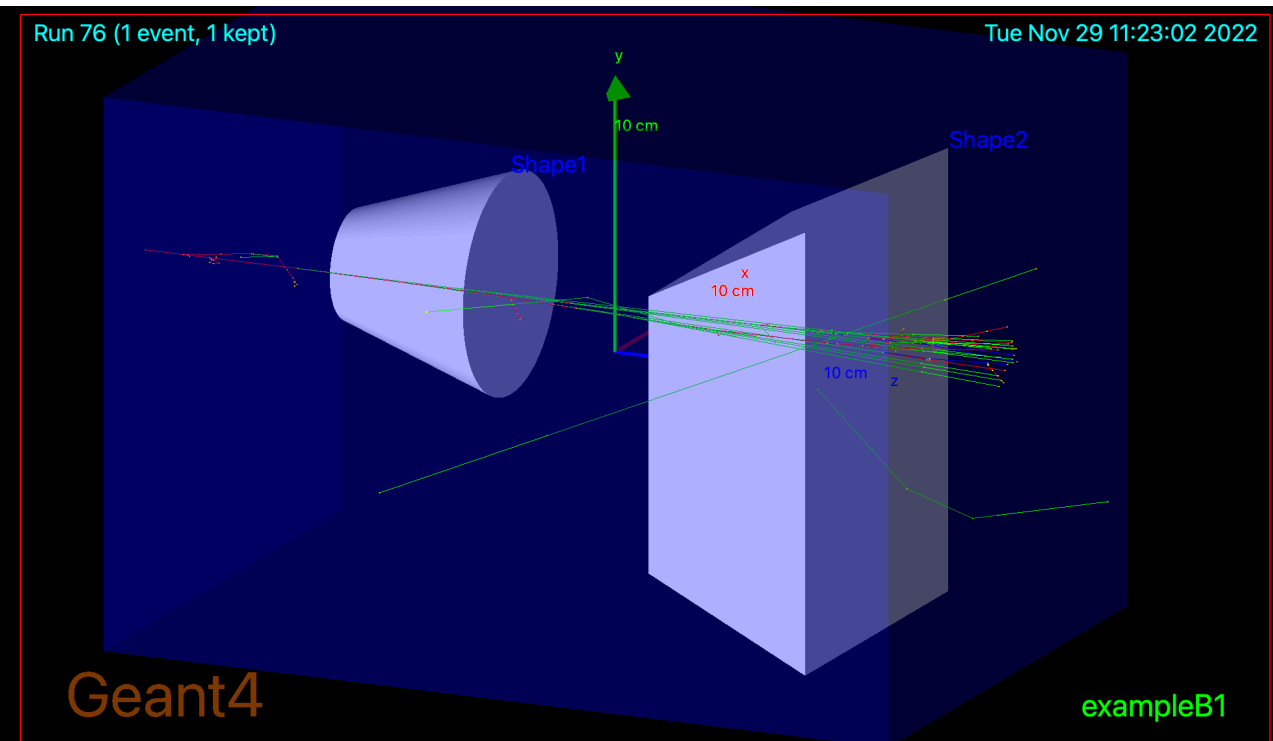
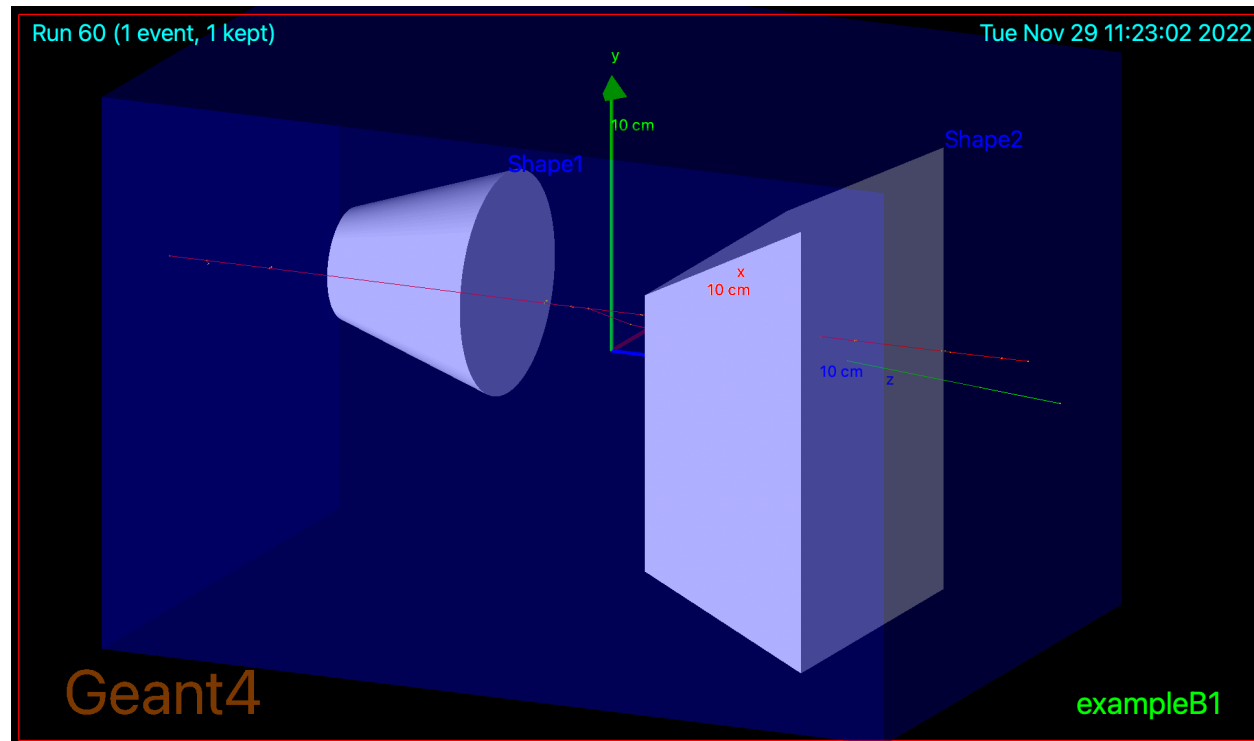


Step #4: Shoot particles

Once you have built any Geant4 application, shooting particles inside and watching the result is *irresistible*

```
/gun/particle mu-  
/gun/energy 1.0 GeV  
/run/beamOn 1
```

```
/gun/particle e-  
/gun/energy 1.0 GeV  
/run/beamOn 1
```



Step #5: Sniff around example B1

(1/4)

Examining code from examples is one of the best ways to learn

- ◆ Example: the main() function

[geant4-11.1.2/examples/basic/B1/exampleB1.cc](#)

```
int main(int argc, char** argv) {
    // some code

    // Construct the default run manager
    //
    auto* runManager =
        G4RunManagerFactory::CreateRunManager(G4RunManagerType::Default);

    // Set mandatory initialization classes
    //
    // Detector construction
    runManager->SetUserInitialization(new DetectorConstruction());

    // Physics list
    G4VModularPhysicsList* physicsList = new QBBC;
    physicsList->SetVerboseLevel(1);
    runManager->SetUserInitialization(physicsList);

    // User action initialization
    runManager->SetUserInitialization(new ActionInitialization());

    // some code
}
```


Step #5: Sniff around example B1

(2/4)

Examining code from examples is one of the best ways to learn

◆ Example: the `main()` function

Every Geant4 application has a *run manager*

There are different run manager *types* (single-threaded vs. multi-threaded)

Both the *physic list* and the *detector geometry* are passed to the run manager

and *user actions* too...

[geant4-11.1.2/examples/basic/B1/exampleB1.cc](#)

```
int main(int argc, char** argv) {
    // some code

    // Construct the default run manager
    //
    auto* runManager =
        G4RunManagerFactory::CreateRunManager(G4RunManagerType::Default);

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    runManager->SetUserInitialization(new DetectorConstruction());

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    G4VModularPhysicsList* physicsList = new QBBC;
    physicsList->SetVerboseLevel(1);
    runManager->SetUserInitialization(physicsList);

    // User action initialization
    runManager->SetUserInitialization(new ActionInitialization());

    // some code
}
```

Step #5: Sniff around example B1

(3/4)

Examining code from examples is one of the best ways to learn

- ◆ Example: the `DetectorConstruction()` class

[geant4-11.1.2/examples/basic/B1/include/DetectorConstruction.hh](#)

```
class DetectorConstruction : public G4VUserDetectorConstruction
{
public:
    DetectorConstruction();
    ~DetectorConstruction() override;

    G4VPhysicalVolume* Construct() override;

    // some code
};
```

[geant4-11.1.2/examples/basic/B1/src/DetectorConstruction.cc](#)

```
G4VPhysicalVolume* DetectorConstruction::Construct()
{
    // some code

    G4Box* solidWorld = // ...

    G4LogicalVolume* logicWorld = // ...

    G4VPhysicalVolume* physWorld = // ...

    // some code
}
```

Step #5: Sniff around example B1

(4/4)

Examining code from examples is one of the best ways to learn

- ◆ Example: the `DetectorConstruction()` class

Users are responsible for creating the simulated geometry

Geant4 provides *virtual* classes to be inherited by user code that *overrides* the virtual methods

Solids, logical volumes and physical volumes are created in the `::Construct()` method

geant4-11.1.2/examples/basic/B1/include/DetectorConstruction.hh

```
class DetectorConstruction : public G4VUserDetectorConstruction
{
public:
    DetectorConstruction();
    ~DetectorConstruction() override;

    G4VPhysicalVolume* Construct() override;

    // some code
};
```

geant4-11.1.2/examples/basic/B1/src/DetectorConstruction.cc

```
G4VPhysicalVolume* DetectorConstruction::Construct()
{
    // some code

    G4Box* solidWorld = // ...

    G4LogicalVolume* logicWorld = // ...

    G4VPhysicalVolume* physWorld = // ...

    // some code
}
```

Step #6: Basic examples

(1/3)

Basic examples demonstrates simple features on simplified geometries (good for learning)

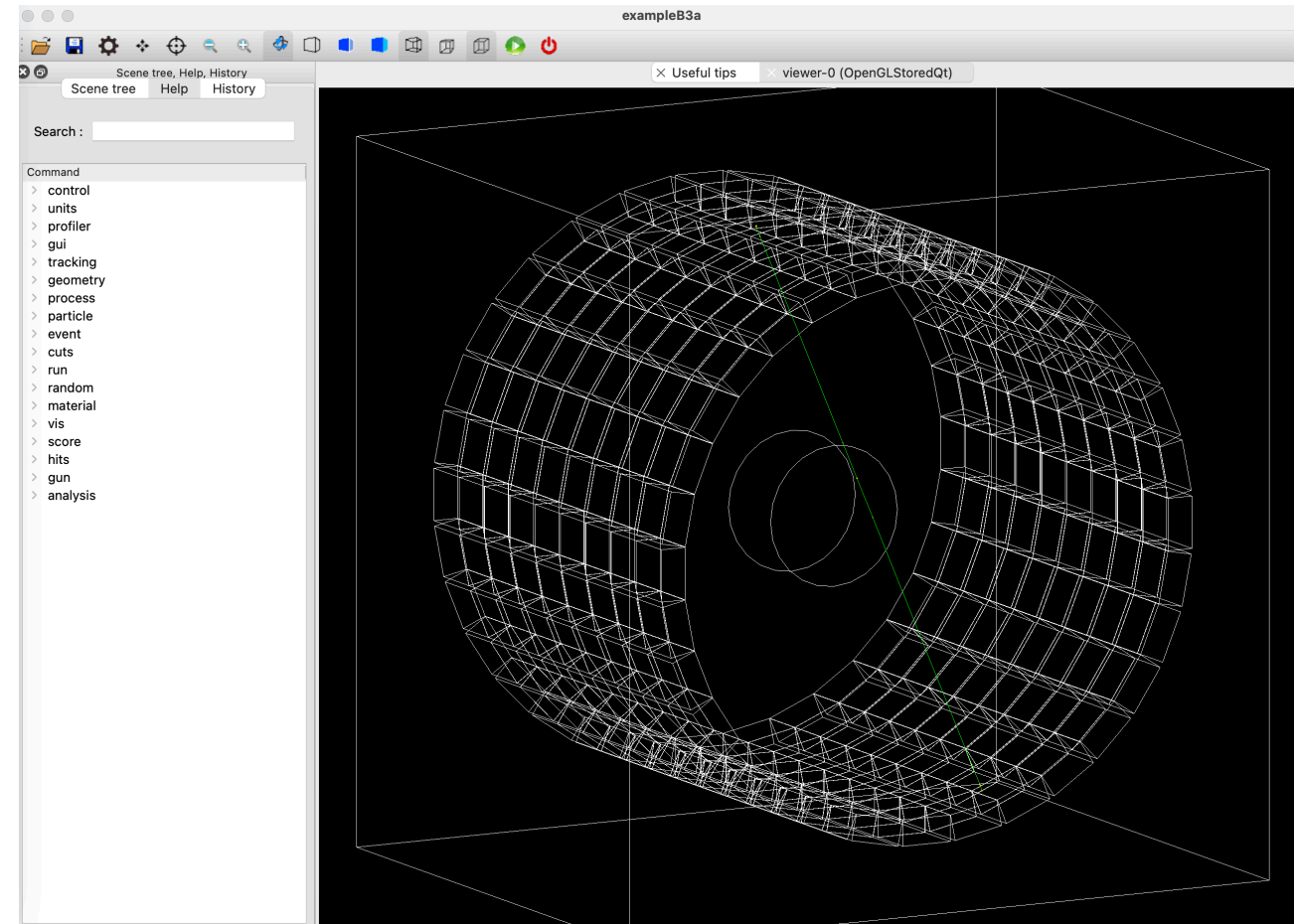
- ♦ **Example B1** (previous slides): simple volumes and scoring with *stepping action*
- ♦ **Example B2**: magnetic field, scoring with *sensitive detectors* and *hits*, *step limiter*

Step #6: Basic examples

(2/3)

Basic examples demonstrates simple features on simplified geometries (good for learning)

- ◆ **Example B1** (previous slides): simple volumes and scoring with *stepping action*
- ◆ **Example B2**: magnetic field, scoring with *sensitive detectors* and *hits*, *step limiter*
- ◆ **Example B3** (PET system): placement with *rotations*, scoring with *scorers*, *radioactive source*



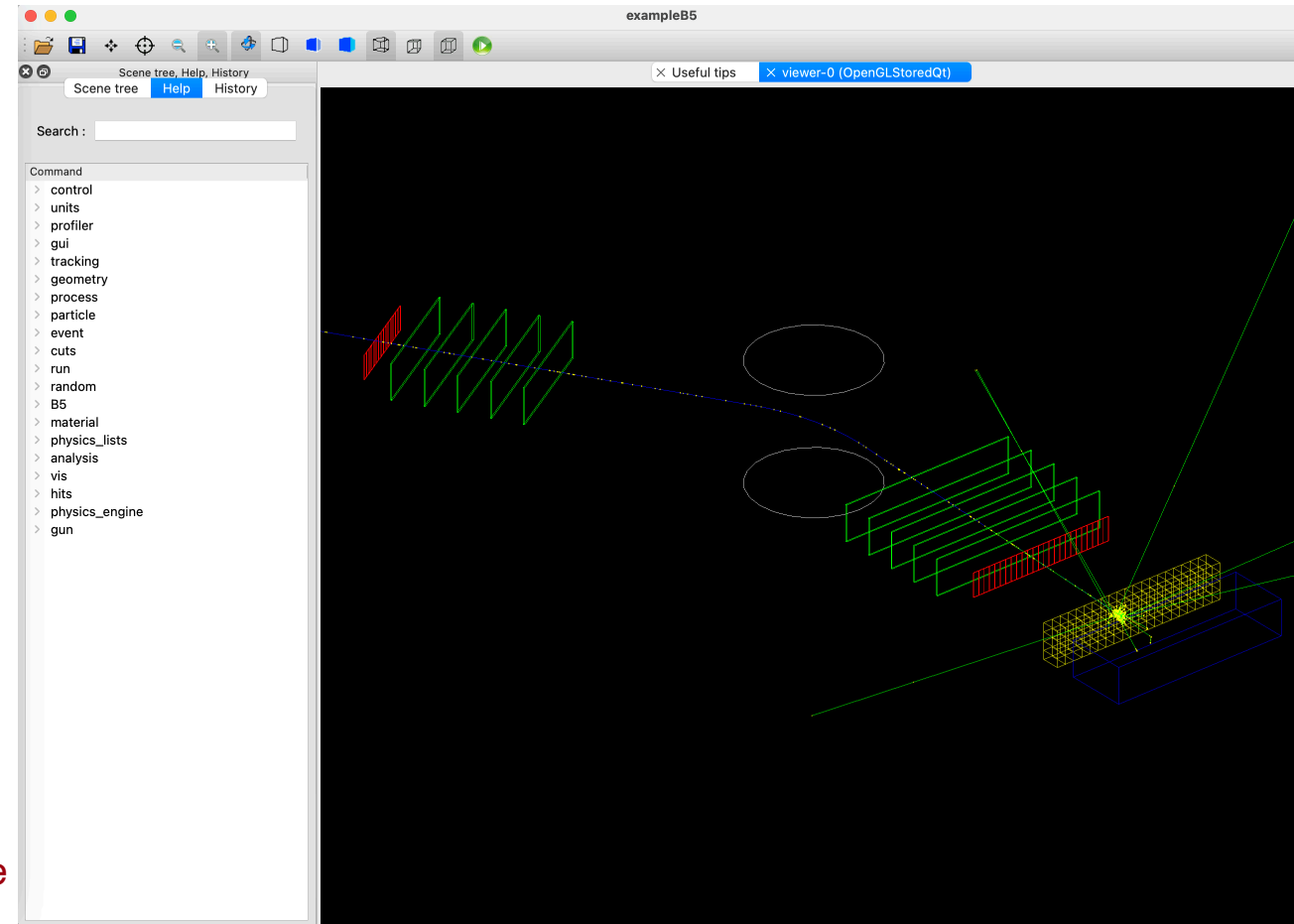
Step #6: Basic examples

(3/3)

Basic examples demonstrates simple features on simplified geometries (good for learning)

- ◆ **Example B1** (previous slides): simple volumes and scoring with *stepping action*
- ◆ **Example B2**: magnetic field, scoring with *sensitive detectors* and *hits*, *step limiter*
- ◆ **Example B3** (PET system): placement with *rotations*, scoring with *scorers*, *radioactive source*
- ◆ **Example B4**: geometry with *replicas*, saving histograms and ntuples with *g4analysis*
- ◆ **Example B5**: *multiple sensitive detectors*, defining *UI commands*, *g4analysis*

equivalent to the
Hands On 2~4



Step #7: Extended and advanced examples

(1/2)

Extended examples demonstrate specific Geant4 features and more complex use cases (some requires external libraries)

- ◆ They are divided in macro areas: *common, eventgenerator, g3tog4, medical, parameterisations, polarisation, visualization, analysis, electromagnetic, exoticphysics, geometry, optical, persistency, radioactive decay, biasing, errorpropagation, field, hadronic, parallel, physicslists, runAndEvent*

Step #7: Extended and advanced examples

(2/2)

Extended examples demonstrate specific Geant4 features and more complex use cases (some requires external libraries)

- ◆ They are divided in macro areas: *common, eventgenerator, g3tog4, medical, parameterisations, polarisation, visualization, analysis, electromagnetic, exoticphysics, geometry, optical, persistency, radioactivedecay, biasing, errorpropagation, field, hadronic, parallel, physicslists, runAndEvent*

Advanced examples demonstrate complex, real-life solutions from domain-specific communities (HEP, biomedical-physics, space science, ...)

- ◆ Each example is a standalone Geant4 simulation targeting a specific application: *doiPET, hadrontherapy, microelectronics, xray_telescope, CaTS, STCyclotron, eRosita, human_phantom, nanobeam, ChargeExchangeMC, air_shower, fastAerosol, iort_therapy, purging_magnet, amsEcal, gammaknife, lAr_calorimeter, radioprotection, HGCal_testbeam, brachytherapy, gammaray_telescope, medical_linac, underground_physics, ICRP110_HumanPhantoms, composite_calorimeter, gorad, microbeam, xray_fluorescence*

Step #8: Refer to the Documentation

The following documents are accessible through the Geant4 website

- ◆ **Book for Application Developers** [[link](#)]: introduces the first-time user to Geant4, provides a description of the available tools and supply the practical information required to develop and run simulation applications
- ◆ **Physics Reference Manual** [[link](#)]: presents the theoretical formulation, model, or parameterization of the physics interactions and describes the probability of the occurrence of an interaction and the sampling mechanisms required to simulate it
- ◆ **Users Guide for Toolkit Developers** [[link](#)]: provides information for those who want to understand or refer to the detailed design of the toolkit, as well as procedures for extending the functionality of the toolkit

Step #9: Refer to the users support channels

Geant4 code can be inspected on

- ◆ **Doxygen** [\[link\]](#): every class and file is available and fully *hyper-linked*
It is useful to overview a class including its *inheritance*

Similarly Geant4 code is accessible through the LXR Code Browser

- ◆ **LXR** [\[link\]](#)

Example: inspect G4Proton class through doxygen

Geant4^{v11.0.3}

Main Page | Namespaces ▾ | **Classes ▾** | Files ▾ | Examples

G4Proton Class Reference

Inheritance diagram for G4Proton:

```
graph BT; G4Proton --> G4Ions; G4Ions --> G4ParticleDefinition;
```

Static Public Member Functions

- static G4Proton * Definition ()
- static G4Proton * ProtonDefinition ()
- static G4Proton * Proton ()

▶ Static Public Member Functions inherited from G4Ions

▶ Static Public Member Functions inherited from G4ParticleDefinition

Static Private Attributes

- static G4Proton * theInstance

Additional Inherited Members

- ▶ Public Types inherited from G4Ions
- ▶ Public Member Functions inherited from G4Ions
- ▶ Public Member Functions inherited from G4ParticleDefinition
- ▶ Protected Types inherited from G4ParticleDefinition
- ▶ Protected Member Functions inherited from G4ParticleDefinition
- ▶ Protected Attributes inherited from G4ParticleDefinition

Step #10: Refer to the forums

Users can reach the Geant4 *community* via the

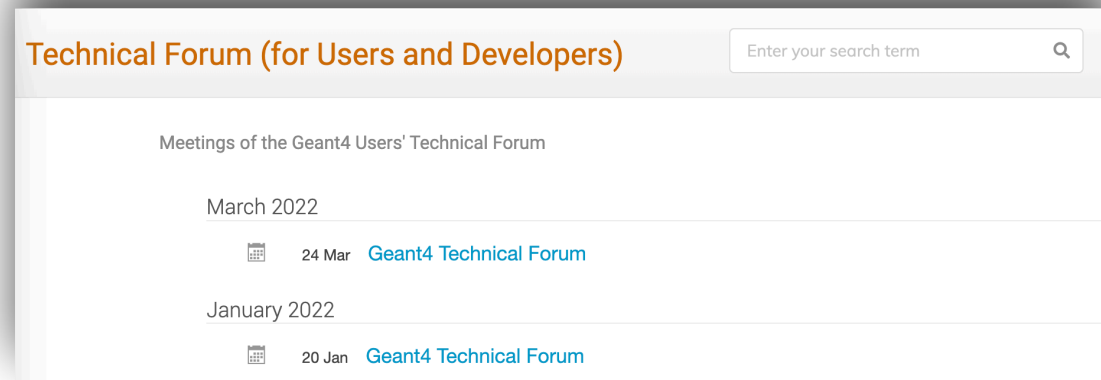
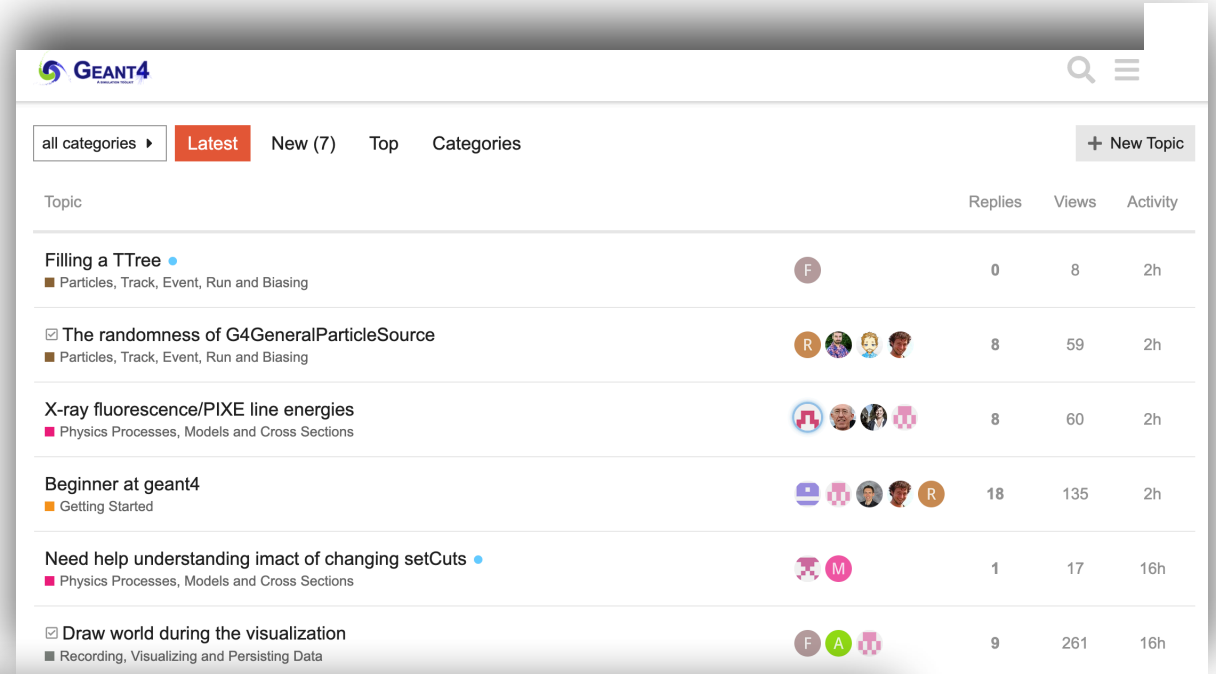
◆ Geant4 Forum [[link](#)]

- ✿ Topics are divided into 9 categories:
forum issues, getting started, geometry fields transportation, physics list, physics processes models and cross sections, particles track event run biasing, recording visualizing and persisting data, ideas and requirements, applications

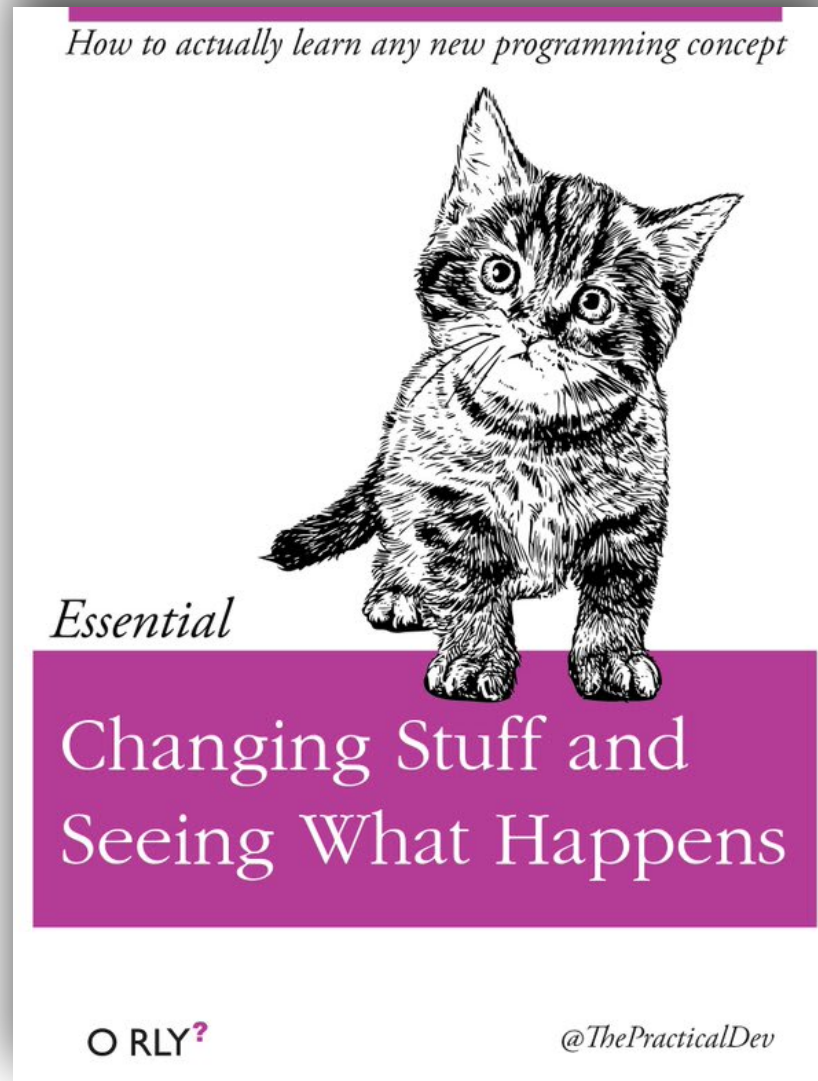
- ✿ Signing-up is required to post a topic

◆ Geant4 Technical Forum [[indico](#)]

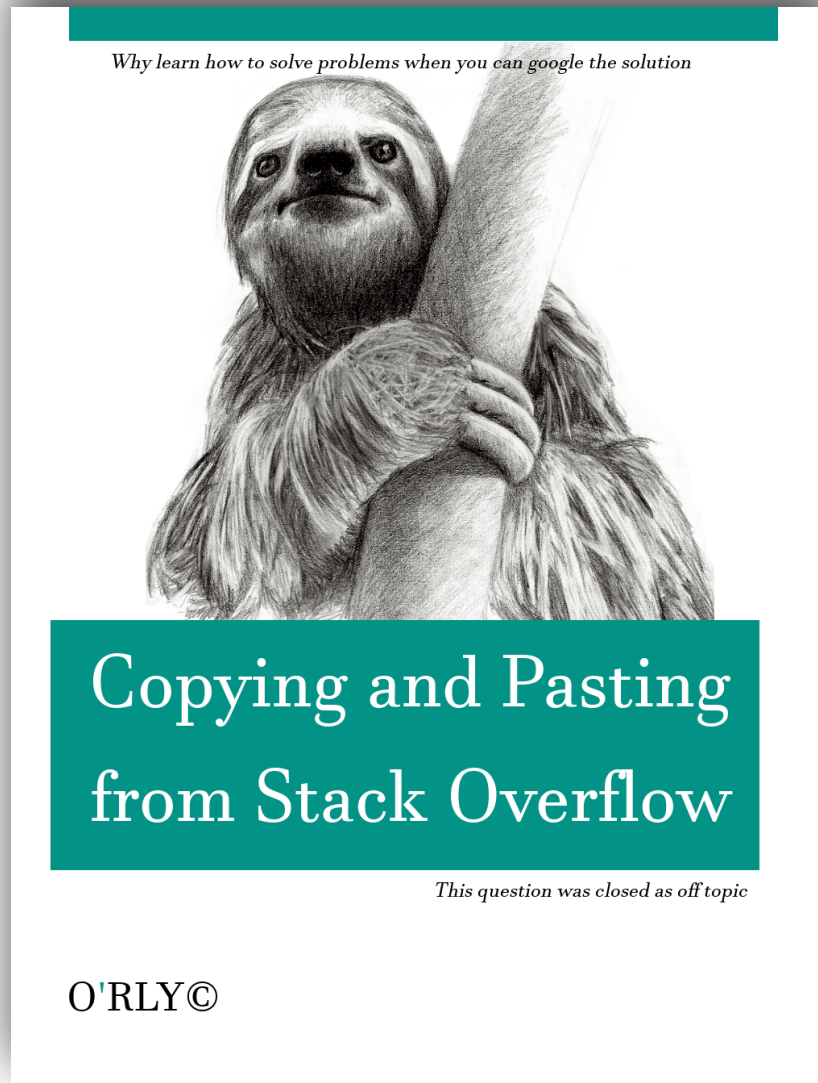
- ✿ Regular meetings between the Geant4 developers and the users community



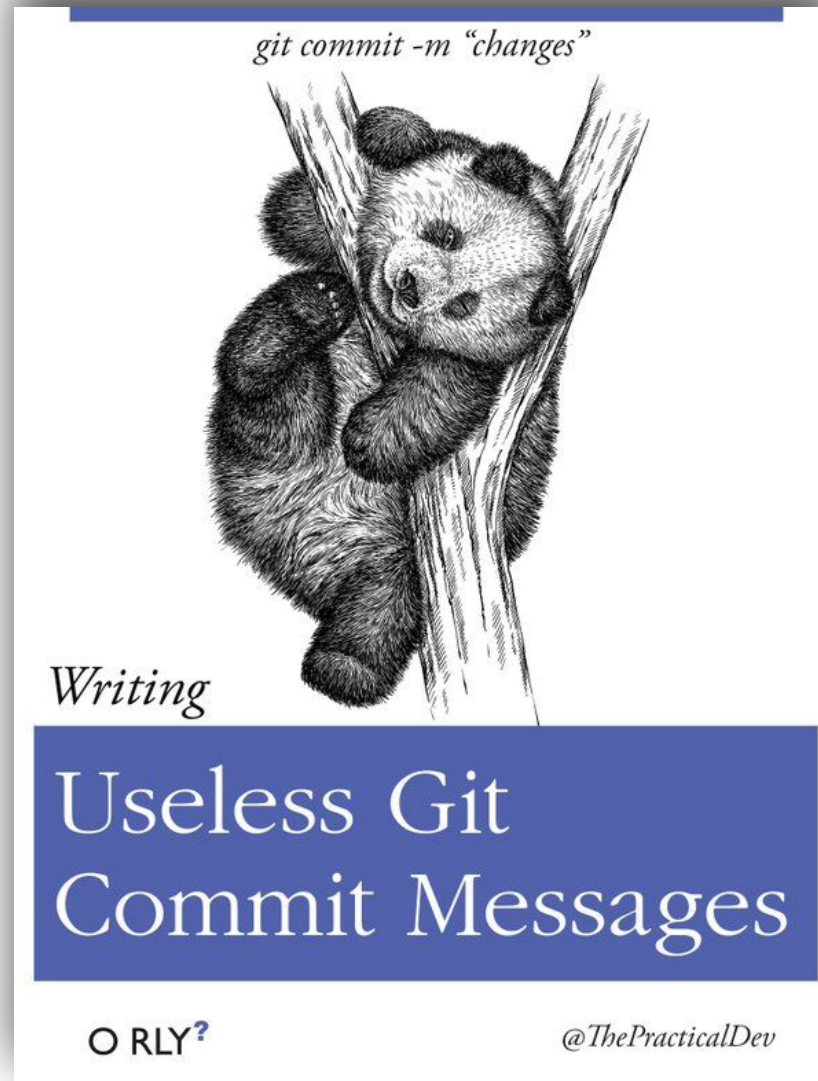
Some recommendations: *how to learn*



Some recommendations: *how to do*



Some recommendations: *try to avoid*



Recap of Examples and Documentation

Geant4 is a modular code made of ~2 million lines of code → *be patient, it takes time to master it*

Few recommended steps:

- ♦ Start from [basic examples](#)
 - ✿ Isolate their building blocks
 - ✿ Adopt the “*change something and see what happens*” mind
- ♦ Inspect Geant4 *classes* on [Doxygen](#)
- ♦ Refer to the [documentation](#) or post your questions on the [user forum](#)